

Session 1 (ASK): Solutions and facilitation notes

Foundations of Clinical Research · Level 1 · Neudata · *Instructor copy. Do not hand out before the exercises.*

These are **model** answers. Many other answers are acceptable: reward work that is specific, consistent and could be acted on. Use the "what good looks like" boxes when you circulate.

Quick quiz (segment 1.1): Rank the evidence

Answer (weakest to strongest): A (expert opinion), then B (registry cohort), then C (randomised trial, CLASSIC), then D (systematic review / meta-analysis of trials).

Source	Design	Real example
A	Expert opinion	None
B	Observational cohort	Similar to Paper 2: Hyun 2023, <i>Ann Intensive Care</i> . Positive cumulative fluid balance on ICU days 2–3 associated with higher 28-day mortality (day 2 cohort HR 2.13, 95% CI 1.48–3.06)
C	Randomised controlled trial	CLASSIC: Meyhoff et al., <i>N Engl J Med</i> 2022. 1,554 ICU adults with septic shock; 90-day mortality 42.3% restrictive vs 42.1% standard
D	Systematic review / meta-analysis	Pooled analyses of fluid-restriction trials

Discussion points

- Why might B and C disagree? In a cohort, sicker patients receive more fluid, so fluid can look harmful even if it is not. This is **confounding by indication** (taught in Session 4). Randomisation removes this.
- What could make C weaker than B? Very small size, many losses to follow-up, poor allocation concealment, an outcome that does not matter to patients, or a population unlike ours.
- The pyramid ranks **designs**, not individual papers. Quality must still be checked (Session 8).

Check your understanding (segment 1.1): types of research

Item	Answer
A Counting ICU pressure injuries this year	Quantitative, primary
B Interviewing families about visiting rules	Qualitative, primary
C Re-analysing 3 years of discharge records	Quantitative, secondary (data already exist; still needs ethics review)
D Hand-hygiene survey + nurse focus groups	Mixed methods, primary

Check your understanding (segment 1.1): research, audit or QI?

Use these if the room wants more examples after the slide.

Project	Answer	Ethics route
Checking whether fluid balance is charted daily, against the unit protocol	Audit	Local/clinical-governance approval; usually no ethics committee
Introducing a daily fluid-balance checklist and refining it in PDSA cycles	Quality improvement	Local approval; ethics review if it adds risk or goes beyond routine care
Randomising patients to restrictive or usual fluids	Research	Ethics committee approval + trial registration before the first patient
Interviewing nurses about why they hesitate to stop fluids	Research (qualitative)	Ethics committee approval (consent, confidentiality)
Reviewing 3 years of ICU records to see if fluid balance is linked to mortality	Research (secondary data, observational)	Ethics committee approval or a formal waiver of consent

Key point: rules differ between institutions. When in doubt, ask the ethics committee **before** starting; journals ask for approval or a waiver at submission.

Check your understanding (segment 1.2): spot the PICO (Exercise 1 warm-up)

P	I	C	O	T
Adults 65+ after hip-fracture surgery	Daily non-drug delirium bundle	Usual care	Delirium (screened daily, e.g. 4AT or CAM)	First 7 days after surgery

Common error: labelling "hip fracture" as the outcome. It has already happened and defines who is eligible (P).

Exercise 1: Model PICO(T) answers

Scenario A: Ventilator-associated pneumonia (experimental)

P	Adults on invasive mechanical ventilation for ≥ 48 h in the ICU
I	Twice-daily nurse checklist for head-of-bed elevation 30–45°
C	Usual care (no checklist)
O	Ventilator-associated pneumonia (using a stated definition)
T	During ventilation, up to ICU day 28

In adults ventilated for at least 48 hours in the ICU, does a twice-daily nurse checklist for head-of-bed elevation, compared with usual care, reduce the incidence of ventilator-associated pneumonia by ICU day 28?

Type: analytical, experimental (could be a before–after study or a cluster trial; designs are Session 3).

FINER watch-point: **F**. VAP definitions vary and VAP is not very common, so the group must state the definition and estimate how many cases occur per year.

Scenario B: Pressure injuries (descriptive first)

The honest first question is descriptive, because nobody knows the size of the problem:

What is the incidence of hospital-acquired pressure injuries (stage 2 or worse) among adults admitted to our medical ward over 6 months?

P	Adults admitted to the medical ward
O	New pressure injury, stage 2 or worse
T	During the admission, over a 6-month period

Type: descriptive. No hypothesis is needed: we **estimate** incidence (with a 95% CI).

A later analytical question: *does 2-hourly vs 4-hourly repositioning reduce new pressure injuries in at-risk adults?*

Scenario C: Delirium (experimental)

P	Adults aged 65+ after hip-fracture surgery
I	Daily multicomponent non-drug delirium-prevention bundle
C	Usual care
O	Incident delirium (screened daily with a validated tool, e.g. 4AT or CAM)
T	During the first 7 postoperative days

In adults aged 65 and over after hip-fracture surgery, does a daily non-drug prevention bundle, compared with usual care, reduce incident delirium in the first 7 postoperative days?

FINER watch-points: **E** (consent from patients who may become confused) and **F** (staff time to deliver the bundle every day).

Scenario D: Stroke door-to-needle time (observational or before–after)

P	Adults with acute ischaemic stroke receiving intravenous thrombolysis in our emergency department
E	Pre-hospital ambulance alert
C	No pre-alert
O	Door-to-needle time (minutes)
T	From hospital arrival to start of thrombolysis

In adults with acute ischaemic stroke who receive thrombolysis in our emergency department, is a pre-hospital ambulance alert associated with a shorter door-to-needle time than no alert?

Type: analytical, observational (PECO). Note: the outcome is continuous (minutes), so the hypothesis is about a difference in median or mean time.

Scenario E: Medication errors (experimental or before–after)

P	Adults transferred from ICU to a general ward
I	Pharmacist-led medication reconciliation at transfer
C	Usual transfer process
O	Patients with at least one medication discrepancy
T	Within 48 hours of transfer

In adults transferred from the ICU to a ward, does pharmacist-led medication reconciliation at transfer, compared with the usual process, reduce the proportion of patients with at least one medication discrepancy within 48 hours?

Watch-point: "medication error" must be defined (discrepancy vs harmful error); one primary outcome only.

What good looks like (Exercise 1)

- All boxes filled; the comparison is named, not "nothing".
- One primary outcome, **measurable**, with a **time point**.
- The sentence reads naturally and contains every PICO element.
- Common faults to correct: two outcomes in one question ("mortality and length of stay"); outcome without time; population too broad ("all patients"); intervention not defined ("better care").

Exercise 2: FINER

Part A: answers vary. Push pairs to justify each colour. A red **F** is the commonest and the most useful to discuss.

Part B: model answers

#	Main problem	Why
1	Feasible	Whole country, 10 years, mortality: far beyond any department's resources. (Large vitamin D trials already exist too, so Novel also fails.)
2	Ethical	Withholding antibiotics in septic shock causes harm; there is no equipoise. An ethics committee would refuse it.
3	Novel	The answer is already measured monthly. This is an audit question rather than a research question, so use the existing data.
4	Relevant (and Interesting)	No plausible mechanism and no impact on practice.

Part B is also an extra slide. Running example (slide): F red (a mortality difference needs well over a thousand patients: CLASSIC 1,554, CLOVERS 1,563); N amber (large trials exist, no mortality difference found); E amber (equipoise exists, but consent in septic shock is difficult); I and R green.

Feasible rewrites: a pilot with cumulative fluid balance at day 5 as the primary outcome (as Paper 1 did); a descriptive study of fluid balance in our unit; an observational study of fluid balance and organ support.

Exercise 3: Aim, objectives and hypothesis

Model answer: delirium example

- *Aim:* To compare a daily non-drug delirium-prevention bundle with usual care in adults aged 65 and over after hip-fracture surgery.
- *Objectives:* 1 (primary) To compare the incidence of delirium in the first 7 postoperative days between groups. 2 To compare length of hospital stay between groups.
- *Hypothesis:* H0: the incidence of delirium in the first 7 days is the same in both groups. H1: it differs.
- *Descriptive version (no hypothesis):* To estimate the proportion of patients aged 65+ who develop delirium in the first 7 days after hip-fracture surgery.

Vague-verb check: rewrite "To understand pressure injuries on our ward" as *"To estimate the incidence of new stage 2 or worse pressure injuries among adults admitted to our ward over 6 months."*

a) Early mobilisation

- H0: ICU length of stay is the **same** with early mobilisation and with usual care.
- H1: ICU length of stay **differs** between early mobilisation and usual care (two-sided).

b) Proportion with positive fluid balance: No hypothesis needed. This is descriptive: we *estimate* a proportion with its 95% confidence interval.

What good looks like

- H0 always states "no difference / no association".
- H1 is usually two-sided ("differs"), even when we expect benefit, because the intervention could also harm.
- The hypothesis uses the **same** outcome and time point as the primary objective.
- Correct the misconception: data can **reject** H0 or **fail to reject** it; we never "prove H0". "No significant difference" is not "no difference" (Paper 1, n = 48, could not detect a mortality difference because it was not powered for it).

Exercise 4: Proposal builder 1 (what good looks like)

A worked model using the **feasible rewrite** of the running example (a pilot). Groups choose their own topics; use this as the quality benchmark, not as the answer. The references, problem statement and title are added next Saturday (Session 2, Exercise 4).

Type of project: research (randomised; changes care), so it needs ethics committee approval and prospective trial registration.

PICO(T): P adults with septic shock within 24 h of starting vasopressors · I restrictive fluid protocol · C usual care · O cumulative fluid balance (ml/kg) · T at day 5.

Question: *In adults with septic shock in our ICU, does a restrictive fluid protocol, compared with usual care, reduce cumulative fluid balance at day 5?*

FINER: F green/amber (pilot of about 50 patients over 12–18 months) · I green · N amber to green (local setting, different protocol) · E amber (deferred or surrogate consent, Session 7) · R green.

Aim: To compare a restrictive fluid protocol with usual care in adults with septic shock in our ICU.

Objectives:

1. (Primary) To compare cumulative fluid balance (ml/kg) at day 5 between groups.
2. To estimate recruitment rate and protocol adherence (feasibility).
3. To describe 28-day mortality and kidney injury needing dialysis in each group (exploratory).

Hypothesis: H0: cumulative fluid balance at day 5 is the same in both groups. H1: it differs.

Marking guide for circulating (not a grade)

Element	Good	Needs work
Topic	One focused problem from the unit	Three problems merged; "improve care in the ICU"
Question	All PICO(T) elements; one primary outcome with time point	Two outcomes; no comparison; vague population
FINER	Each letter justified; red letters fixed	Ticked without thought
Research / audit / QI	Correctly classified; approval route noted	"It's only an audit" for a randomised project
Aim/objectives	One aim; 2–3 measurable objectives; primary marked	"To understand...", "to study..."; more than 4 objectives
Hypothesis	Matches primary objective; H0 and H1; or correctly "descriptive"	Hypothesis about a different outcome

Exercise 5: Synonym list (homework)

Model for the running example: **P** septic shock · sepsis · vasopressor-dependent sepsis; **I** fluid restriction · restrictive fluid · conservative fluid · de-resuscitation; **O** mortality · death · survival. For the pressure-injury example: pressure ulcer · pressure injury · bed sore · decubitus ulcer; repositioning · turning · position change.

Exit question: answers

1. **Feasible.** A mortality outcome needs well over a thousand patients (Novel is also amber because large trials already exist).
2. e.g. *"To estimate the incidence of new stage 2 or worse pressure injuries among adults on our ward over 6 months."*